

DUL-104 — Current/Live Demonstration Knowledge Pack

Synthetic current-well record used to demonstrate NWIS anomaly detection + historical retrieval

IMPORTANT: This document is synthetic demonstration data prepared for the NWIS/SIH prototype. It must not be represented as an authentic confidential OIL record, real incident report, or verified SPE publication.

Well Summary

Field	Value
Well	Duliajan-104 (DUL-104)
Report date	05-Dec-2023
Current depth	2205 m MD
24-hour progress	85 m
Formation	Barail Group — Upper Shale
Current ROP	14.5 m/hr
Mud weight	1.20 g/cc
Torque	18.2 kN·m

24-Hour Narrative

00:00–08:00: drilled from 2120 m to 2165 m in Upper Barail sandstone with stable parameters. 08:00–16:00: transitioned into Barail Reactive Shale at 2185 m; surface torque rose from 11.5 to 18.2 kN·m; 12-tonne overpull was observed during a connection near 2200 m. 16:00–24:00: proactive back-reaming was initiated, a 25 bbl high-viscosity sweep was pumped, and mud was conditioned with 2% Poly-Glycol. A proactive alert was triggered based on similarity to the historical DUL-92 signature.

Live Feature Vector for ML

Feature	Current value
Depth	2205 m
Torque	18.2 kN·m
ROP	14.5 m/hr
RPM	120 rpm (synthetic live-stream value)
WOB	18 kN (synthetic live-stream value)
Standpipe pressure	2400 psi (synthetic live-stream value)
Mud weight	1.20 g/cc
Formation	Barail Reactive Shale

How NWIS Uses This Record

1. GIS identifies relevant offset wells such as DUL-92. 2. The anomaly model checks whether the live sensor pattern is unusual. 3. RAG searches historical reports for similar formation/depth/event patterns. 4. The LLM explains the similarity and cites the historical report. 5. The dashboard shows the warning, trend, map, and evidence.

Example AI Output

Potential Barail reactive-shale / stuck-pipe risk. Current torque is elevated at 2205 m after entering Barail Shale. A historical DUL-92 case occurred around 2210 m. The system should present the historical mitigation as

evidence and require engineer confirmation before any operational action.

Important ML Note

This PDF is not sufficient to train a robust anomaly model. The prototype should pair it with a CSV/time-series dataset containing many timestamped observations from multiple synthetic wells, including normal and anomalous intervals.